

Systems Verification

Exercise Sheet 07 - Solutions

Exercise 1: Verification of Toastmasters Pro Microwave

You are part of a very old-school company that specializes in verifying systems of various kitchen appliances by hand without using the help of systems like SPIN (or smart stuff like common sense) when you get the following job by a start-up focussing on microwaves:

Dear Verificator,

My name is Timmy Toastman and I am the CEO of "ToastMaster Pro intl.". We would like you to verify our new revolutionary system of a microwave. These are the features of our microwave:

- The door can be either open or closed
- The microwave is either turned on or off
- If the door is closed, you can open the door
- If the door is open, you can close the door
- If the microwave is turned on, you can turn it off
- If the microwave is turned off, you can turn it on

Before we start the very laborous process of constructing a machine with these features, we would like you to make sure that the following property holds:

The microwave shouldn't be turned on while the door is open.

With the working contract already signed by your supervisor, feel free to immediately start this verification job!

Yours sincerely,
Timmy Toastman

"Toast the Competition: A Recipe for Success!"

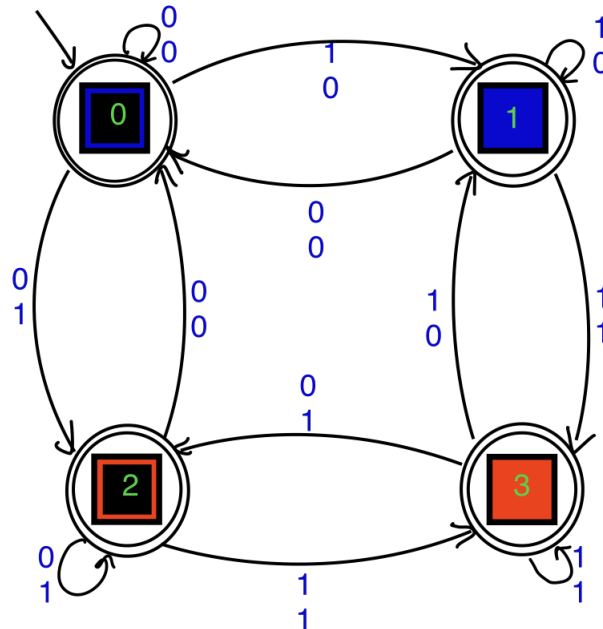
Here is a check-list for you to start this verification job:

1. Think of what logical variables you can use and what they describe.

Solution: To model the behaviour of the microwave described above one needs two boolean variables:

- The variable *a* has the value *True* when the door is open or *False* when it is closed.
 - The variable *b* has the value *True* when the microwave is turned on or *False* when it is turned off.
2. Model the described system as a Büchi automaton. Take the described features very literal (even if you find them suboptimal).

Solution: In the initial state (state 0), the microwave is turned off and the door is closed. From there the automaton can take the transition (1,0) to state 1 in which the microwave is off and the door is open. Alternatively state 2 is also reachable from state 0 via the transition (0,1) in which the microwave is turned on and the door is closed. State 3, in which the microwave is turned on and the door is open, can be reached from state 1 and 2.



3. Transform their desired property into an LTL formula.

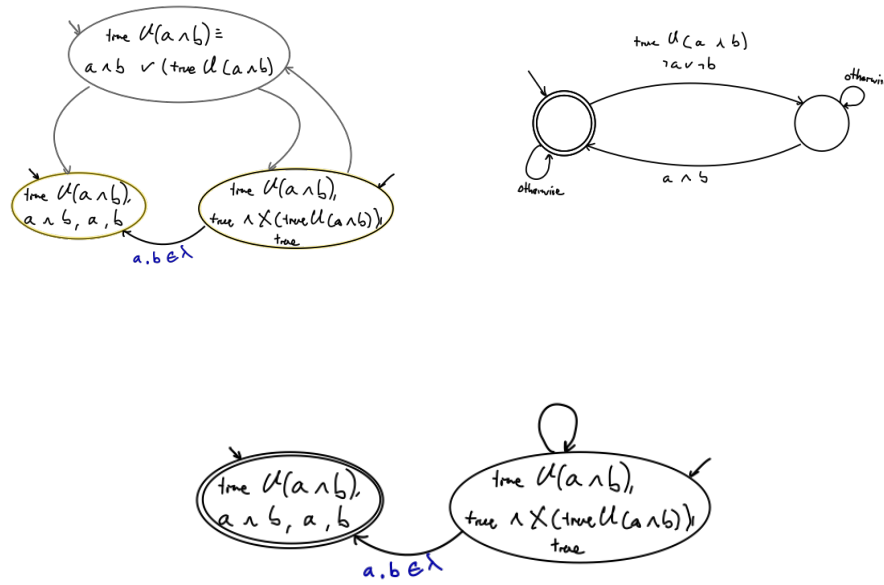
Solution: $\Box \neg (a \wedge b)$.

4. Verify if the model satisfies the desired property. (using the Büchi automata strategies we saw in class)

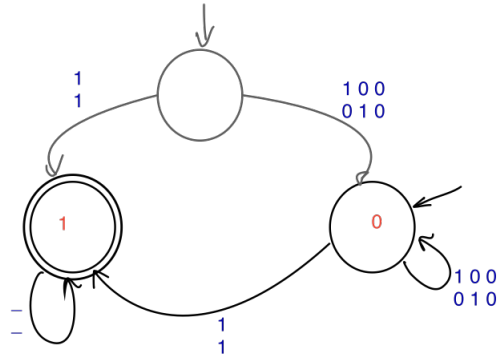
Solution: The general procedure is to build a Büchi automata of the negated LTL formula and to check whether the intersection between it and the automata of our model is empty. If the intersection is empty then our model satisfies the LTL formula otherwise we have a counterexample.

(a) Negate the LTL formula: $\neg(\Box \neg (a \wedge b)) \equiv \Diamond \neg \neg (a \wedge b) \equiv \Diamond (a \wedge b) \equiv true \mathcal{U} (a \wedge b)$

(b) Build the SLBA:

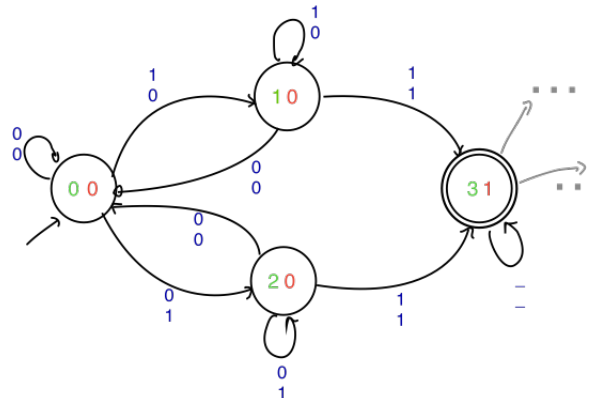


(c) Build the Büchi automata from the SLBA:



To simplify the following step, we optimize the automata by removing the grey parts and making state 0 an initial state. This smaller automata accepted the same language and is equivalent.

(d) Intersect the Büchi automata of our system with the Büchi automata of the negated LTL formula:



The intersection is created by starting with an initial state that is labelled according to the initial states of our intersecting automata (in our case a green and red 0). We then create a target state for each transition that is possible in both automata. The target states are also labeled according to the target states in the intersecting automata. For example by reading a $(0, 1)$ in state 0 of our original model we reach state 2 (green label) while in state 0 of the modeled LTL formula we stay in state 0 (red label). Therefore we create the state labeled with a green 2 and a red 0 and add the transition $(0, 1)$ between it and the initial state. States are labeled accepting, if the respective run passes an accepting state in **both** intersecting models.

5. If their modelled system fails, give a better model for their microwave and show why your model doesn't fail.

Solution: We simply remove state 3 of our original automaton and all the transitions that go to and from it:

